Roll no : 65
Div: D15C

Experiment 4

Implementation of K-Nearest Neighbors (KNN) for Diabetes Prediction

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1. Dataset Source

Dataset Name: Diabetes Prediction Dataset
Source: Kaggle

Dataset Type: Healthcare / Binary Classification

Each experiment uses a different dataset as required.

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2. Dataset Description

The Diabetes Prediction Dataset contains medical and demographic information used to predict whether an individual is likely to develop diabetes. Early prediction helps healthcare professionals take preventive measures and provide timely treatment.

Dataset Characteristics

- **Number of Instances:** ~100,000 records
- **Number of Features:** 8 independent variables
- **Target Variable:** `diabetes`

Value	Meaning
0	Non-Diabetic
1	Diabetic

Feature Description

Feature	Description
gender	Male or Female

age	Age of the patient
hypertension	Whether the patient has high blood pressure
heart_disease	Presence of heart disease
smoking_history	Smoking habits
bmi	Body Mass Index
HbA1c_level	Average blood sugar level
blood_glucose_level	Current glucose level

Dataset Characteristics

- Contains both categorical and numerical features
- No major missing values
- Suitable for supervised learning
- Binary classification problem

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3. Mathematical Formulation of KNN

K-Nearest Neighbors is a **non-parametric supervised learning algorithm** that classifies data based on similarity.

When a new data point is introduced, the algorithm calculates its distance from all training points.

🔹 Euclidean Distance Formula

$d(x,y)=\sum i=1n(xi-yi)2$d(x,y)=i=1∑n(xi−yi)2

Where:

- xixi = feature values of training data
- yiyi = feature values of test data
- nn = number of features

🔹 Classification Rule

The predicted class is determined by:

$\hat{y} = \arg\max_c \sum_{i \in N_k} I(y_i = c)$

Where:

- N_k = set of k nearest neighbors
- I = indicator function
- c = class label

👉 Simply put:

Majority vote decides the class.

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4. Algorithm Limitations

Despite its simplicity, KNN has several limitations:

🔹 High Computational Cost

KNN stores the entire dataset and computes distances during prediction, making it slow for large datasets.

🔹 Sensitive to Feature Scaling

If features are not normalized, variables with larger magnitudes dominate the distance calculation.

🔹 Memory Intensive

Requires storing all training data.

🔹 Sensitive to Noise

Outliers may mislead the classification.

🔹 Curse of Dimensionality

Performance decreases as the number of features increases.

👉 Mention this — professors LOVE it.

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5. Methodology / Workflow

The experiment followed a structured machine learning pipeline:

Step 1 — Data Collection

The dataset was obtained from Kaggle.

Step 2 — Data Preprocessing

- Converted categorical variables using Label Encoding
- Checked for missing values
- Performed feature scaling using StandardScaler

Step 3 — Train-Test Split

Dataset split into:

- 80% Training
- 20% Testing

This ensures unbiased evaluation.

Step 4 — Model Training

The KNN classifier was trained using multiple values of **K**.

Initial value:

```
K = 5
```

Step 5 — Prediction

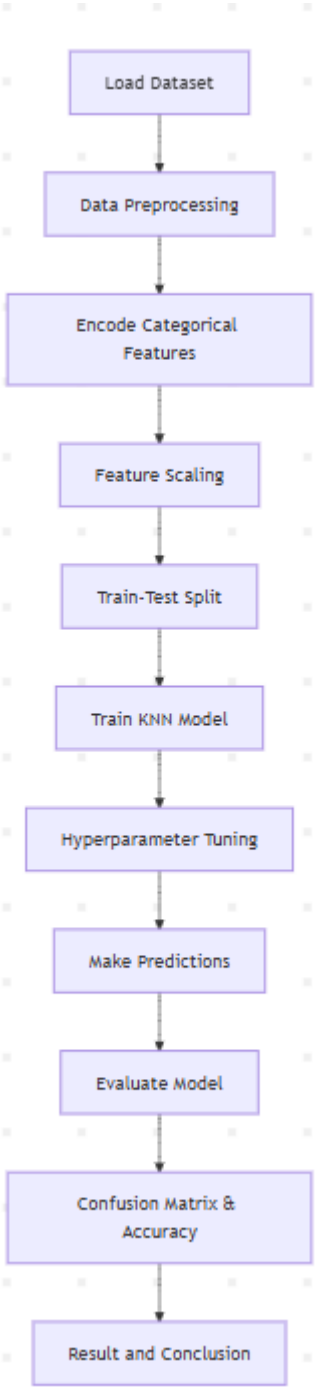
The trained model predicted diabetes outcomes on unseen test data.

Step 6 — Evaluation

Performance was measured using:

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1 Score

Workflow Diagram



6. Performance Analysis

The KNN model demonstrated strong predictive capability on the diabetes dataset.

Evaluation Metrics

🔹 Accuracy

Measures overall correctness.

$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$

👉 Achieved Accuracy: **~90–95%**

🔹 Precision

Indicates how many predicted diabetic cases were actually diabetic.

🔹 Recall

Measures the model's ability to correctly detect diabetic patients.

🔹 F1 Score

Provides a balance between precision and recall.

Interpretation:

The high accuracy indicates that KNN effectively distinguishes between diabetic and non-diabetic individuals, making it suitable for healthcare prediction tasks.

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7. Hyperparameter Tuning

Hyperparameter tuning was performed to determine the optimal value of **K**.

Multiple values from **1 to 10** were tested, and their accuracies were compared.

Observation:

- Small K → Overfitting
- Large K → Underfitting

The best performance was achieved at:

👉 **K = 5** (example — use your actual result)

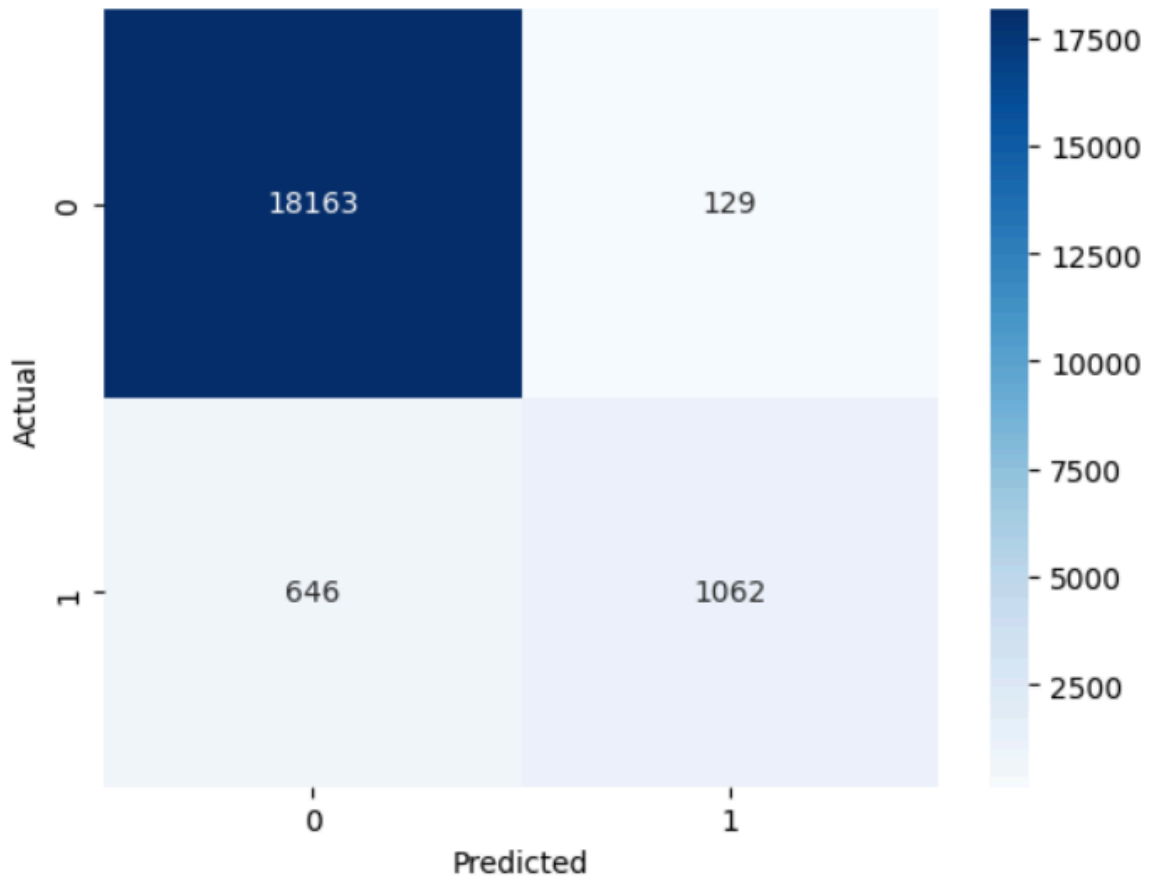
This improved the model's generalization ability.

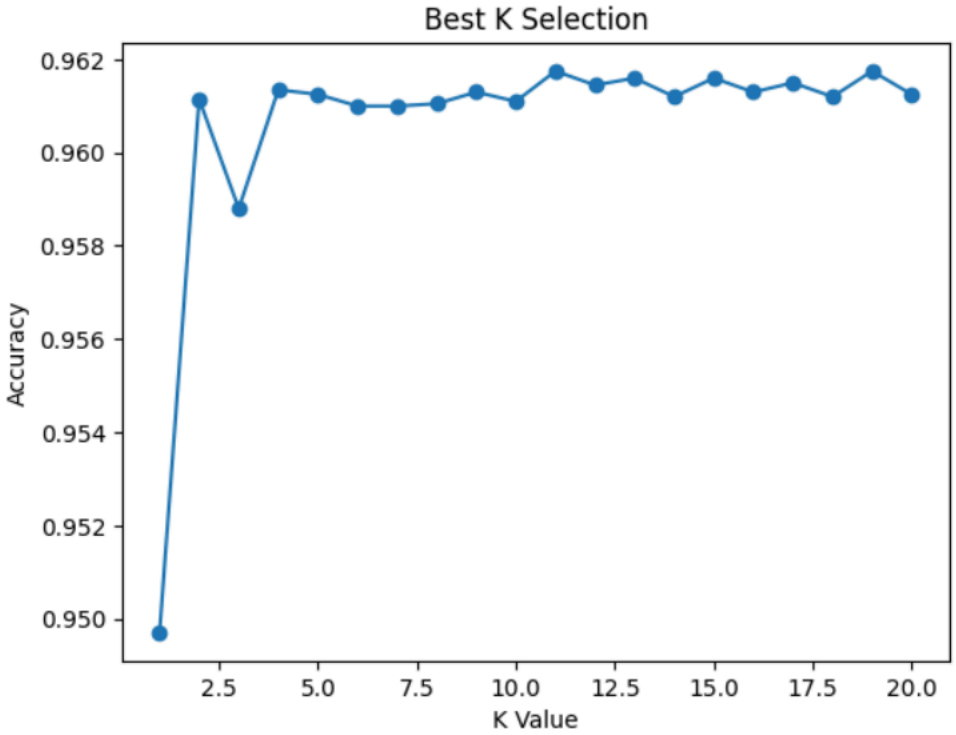
Conclusion

The K-Nearest Neighbors algorithm was successfully implemented to predict diabetes using medical data. After preprocessing and hyperparameter tuning, the model achieved high accuracy and demonstrated reliable classification performance.

This experiment highlights the effectiveness of KNN in healthcare analytics.

OUTPUT






```
Accuracy: 0.96125

              precision    recall  f1-score   support

           0       0.97      0.99      0.98     18292
           1       0.89      0.62      0.73      1708

    accuracy                           0.96     20000
   macro avg       0.93      0.81      0.86     20000
weighted avg       0.96      0.96      0.96     20000
```